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VECTOR SPACE PROBLEM SHEET

This sheet reinforces the lectures on vector spaces and inner products. Question 1 uses actual numbers; the later questions guide you, one step at a time, through proving some well known results. The algebra in each step is short. Try to explain each step in words as well as symbols: a good proof is one that someone else can follow without having to fill in gaps.

This looks very mathematical, but it helps to develop the tools and language that are used to describe machine learning.

Throughout, an inner product $\langle x, y \rangle$ between vectors in a vector space $\mathcal{V}$ satisfies the following properties

1. $\langle x, x \rangle \geq 0$ for all $x \in \mathcal{V}$
2. $\langle x, x \rangle = 0$ if and only if $x = 0$
3. $\langle \alpha x, y \rangle = \alpha \langle x, y \rangle$
4. $\langle x, y + z \rangle = \langle x, y \rangle + \langle x, z \rangle$
5. $\langle x, y \rangle = \langle y, x \rangle$

For vectors in $\mathbb{R}^n$ the usual inner product is $\langle x, y \rangle = x^\top y = \sum_{i=1}^n x_i y_i$, with norm $\|x\| = \sqrt{\langle x, x \rangle}$.

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1 This question is a warm-up using actual numbers. Let

$$\mathbf{x} = \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix}, \quad \mathbf{y} = \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix}$$

with the usual inner product $\langle \mathbf{x}, \mathbf{y} \rangle = \mathbf{x}^\top \mathbf{y}$ on $\mathbb{R}^3$.

- (a) Compute $\langle \mathbf{x}, \mathbf{y} \rangle$, the norms $\|\mathbf{x}\|$ and $\|\mathbf{y}\|$, and the distance $d(\mathbf{x}, \mathbf{y}) = \|\mathbf{x} - \mathbf{y}\|$.
[4 marks]

4

- (b) Using $\langle \mathbf{x}, \mathbf{y} \rangle = \|\mathbf{x}\| \|\mathbf{y}\| \cos(\theta)$, find the angle θ between $\mathbf{x}$ and $\mathbf{y}$. Check that the Cauchy-Schwarz inequality, $|\langle \mathbf{x}, \mathbf{y} \rangle| \leq \|\mathbf{x}\| \|\mathbf{y}\|$, and the triangular inequality, $\|\mathbf{x} + \mathbf{y}\| \leq \|\mathbf{x}\| + \|\mathbf{y}\|$, both hold for these two vectors.
[5 marks]

5

(c) The projection of y onto x is

$$\text{proj}_x(y) = \frac{\langle y, x \rangle}{\langle x, x \rangle} x.$$

Compute it, and the remainder $r = y - \text{proj}_x(y)$. Show that r is orthogonal to x . [5 marks]

15

(d) Not every formula that looks like an inner product is one. For $u, v \in \mathbb{R}^2$ consider

$$\langle u, v \rangle = u_1 v_1 - u_2 v_2.$$

Find a vector u with $\langle u, u \rangle < 0$, and a non-zero vector with $\langle u, u \rangle = 0$. Which of the five properties fail? Why does this stop us defining a norm $\|u\| = \sqrt{\langle u, u \rangle}$?
[6 marks]

6

End of question 1

(a) $\frac{4}{5}$ (b) $\frac{5}{5}$ (c) $\frac{5}{5}$ (d) $\frac{6}{6}$ Total $\frac{20}{20}$

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2 This question uses the properties of an inner product to prove two results that hold in *every* inner-product space.

(a) Consider the quadratic

$$q(t) = \langle x + ty, x + ty \rangle.$$

Use properties 3, 4 and 5 to expand $q(t)$ in the form $q(t) = At^2 + 2Bt + C$, and write down A , B and C as inner products. [4 marks]

4

- (b) By properties 1 and 2, $q(t)$ is never negative. A quadratic that is never negative cannot have two distinct real roots (it would be negative between them), so its roots are either complex or a double root. Use the formula for the roots of $At^2 + 2Bt + C = 0$ to show that this implies

$$\langle x, y \rangle^2 \leq \langle x, x \rangle \langle y, y \rangle.$$

This is the famous Cauchy-Schwarz inequality written in a very general form. [6 marks]

This image shows a blank sheet of white paper with horizontal ruling lines. The lines are evenly spaced and extend across the width of the page. There is no handwriting or other markings on the paper.

- (c) From an inner product we can define a norm $\|x\| = \sqrt{\langle x, x \rangle}$. This clearly satisfies non-negativity and “linearity” ($\|ax\| = |a|\|x\|$). The only non-trivial property to show is that this norm satisfies the triangular inequality. Expand out $\|x + y\|^2$ and hence show that

$$\|x + y\|^2 = \|x\|^2 + \|y\|^2 + 2\langle x, y \rangle$$

Then use the Cauchy-Schwarz inequality to prove the triangular inequality. [10 marks]

This image shows a blank sheet of white paper with horizontal ruling lines. The lines are evenly spaced and extend across the width of the page. There is no handwriting or other markings on the paper.

End of question 2

(a) $\frac{4}{4}$	(b) $\frac{6}{6}$	(c) $\frac{10}{10}$	Total $\frac{20}{20}$
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3 Random variables, X , Y , etc. form a vector space (i.e. they satisfy properties such as closure under addition and scalar multiplication). Furthermore we can define an inner product between random variables as

$$\langle X, Y \rangle = \mathbb{E}[XY]$$

(i.e. the expectation of the random variable $Z = XY$)

(a) Use the Cauchy-Schwarz inequality to show that

$$\text{Cov}(X, Y)^2 \leq \text{Var}(X)\text{Var}(Y)$$

hence show that the Pearson correlation is between -1 and 1 . [10 marks]

10

(b) Show that for vectors $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ that the inner product

$$\langle \mathbf{x}, \mathbf{y} \rangle = \sum_{i=1}^n w_i x_i y_i$$

satisfies the conditions of an inner product provided $w_i > 0$ for all i . Write down the norm and distance induced by this inner product and provide an interpretation of what this distance means.

[10 marks]

10

End of question 3

(a) $\frac{\quad}{10}$	(b) $\frac{\quad}{10}$	Total $\frac{\quad}{20}$
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4 In the previous question you showed that $\langle \mathbf{x}, \mathbf{y} \rangle = \sum_{i=1}^n w_i x_i y_i$ is an inner product provided every weight w_i is positive. Functions form a vector space in just the same way (we can add them and multiply them by scalars), and if we replace the sum over components by an integral over x we get

$$\langle f, g \rangle = \int_{-\infty}^{\infty} w(x) f(x) g(x) dx$$

which, by the same argument, is an inner product provided the *weight function* $w(x)$ is positive.

In this question we choose $w(x)$ to be the density of a standard normal deviate

$$w(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$$

so that $\langle f, g \rangle = \mathbb{E}[f(X)g(X)]$ where $X \sim \mathcal{N}(0,1)$. That is, this is exactly the inner product of the previous question, applied to functions of a random variable.

The monomials $f_i(x) = x^i$ span the polynomials, but they are not orthogonal. We can orthogonalise them with the Gram–Schmidt procedure: having found $p_0, p_1, \dots, p_{n-1}$ we subtract from f_n its component along each of them,

$$p_n(x) = f_n(x) - \sum_{i=0}^{n-1} \frac{\langle f_n, p_i \rangle}{\langle p_i, p_i \rangle} p_i(x), \quad p_0(x) = 1.$$

Orthogonal polynomials are useful because they make approximating a function easy. If we want the best polynomial approximation to some function f we can compute each coefficient separately with a single inner product, and if we later decide to add a higher order term none of the coefficients we have already found change. Using the monomials instead we would have to solve a set of simultaneous equations, and start again from scratch each time we added a term.

You will need the integrals below, which you are not expected to compute. (They are the moments of a standard normal deviate. All the odd moments vanish because $w(x)x^n$ is an odd function.)

$\mathbb{E}[X^n] = \int_{-\infty}^{\infty} w(x)x^n dx$	0	1	2	3	4	5	6
1	0	1	0	3	0	15	

(c) Find $p_3(x)$.

[5 marks]

5

